

04 Rate Laws

Background for this lecture:

- Chemical reactor design is the process of designing reactor systems around chemical kinetics.
- As we study CRD, we aim to see how different types of kinetic scenarios influence reactor design decisions.
- Derivation of rate expressions is not at all straightforward ... Doing this for even one reaction could extend over multiple PhD dissertations or decades of industrial R&D work.
- However, often, simplified forms of rate expressions provide nice estimates for reactor design. Hence, we use simplified expressions in this class, and also in Process Design.
- We will get into some more detailed rate expressions later in this class (when we discuss mechanisms) ... For now, the goal is to be able to construct a reasonable rate expression that can be used in a CRE/CRD problem.

A rate law is a model

They are used to model how the net rate of generation of a species changes due to changes in

concentration and temperature

Often, the changes in concentration for a chemical reaction can be modeled using

a power law model

For example, take the reaction $A + B \rightarrow C + D$. Assuming this reaction is irreversible, a good way to model the concentration dependence of the rate law is

$$-r_A = k C_A^n C_B^m$$

where n & m are

exponents

Theoretically, n & m could span from $-\infty$ to $+\infty$

They don't have to be positive, they don't have to be integers

In this class, we are often concerned with two specific cases:

- First specific case: The reaction is elementary.

This means that the rate law is $-r_A = k_A(T) \cdot \prod_{i=1}^{\text{\#reactants}} C_i^{r_i}$

product of the concentrations of the participating species raised to the power of their stoichiometric coefficients

Therefore, for the irreversible reaction $A + B \rightarrow C + D$, $-r_A =$

$$-r_A = k_A C_A C_B$$

Let's do another example. The rate law for the reaction $2NO + O_2 \rightarrow 2NO_2$ is elementary. What is the rate expression?

$$-r_{NO} = k_{NO} C_{NO}^2 C_{O_2}$$

Two important notes pertaining to elementary reactions:

- One: If you are told that a reaction is elementary, take that to mean that the reaction is *elementary as written*. Do not alter the stoichiometry for the reaction and then write the rate expression based off the altered stoichiometry. For example, it would be incorrect to alter the stoichiometry of the reaction above to $NO + 1/2 O_2 \rightarrow NO_2$ and then write the rate law as $-r_{NO} = k_{NO} C_{NO} C_{O_2}^{1/2}$. Note that this gives entirely different concentration dependence of the rate than the expression above. If you are given a reaction and told it's elementary, it means that it's elementary as written.
 - Two: If you are working homework/exam problems in this class and you aren't given information about the rate law, you can assume it's elementary. In this case, write: "assume the rate law is elementary" on your solution and proceed with the solution using the elementary rate expression.
- Second specific case: The reaction is "Nth order".

Sometimes, you'll be told: "The reaction is N order in A" or simply, "The reaction is N

order." In this case:

$$-r_A = k_A \cdot \prod_{i=1}^{\text{\# relevant species}} C_i^{n_i} \quad n_i \text{ reaction orders, } \sum_i n_i = N$$

Often, when the kinetic information is given in this way, it means the rate expression only depends on the concentration of one component. For example:

- $A + B \rightarrow C$. The reaction is 1st order in A means

$$-r_A = k_A C_A$$

- What about this reaction: $\text{CH}_3\text{OCH}_3 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{OHCH}_2\text{OH}$. The reaction is 1st order. What is the rate law? Why?

$$-r_{\text{CH}_3\text{OCH}_3} = k C_{\text{CH}_3\text{OCH}_3}$$

I'm going to assume that $C_{\text{H}_2\text{O}} \gg C_{\text{CH}_3\text{OCH}_3}$ so the kinetics are controlled much more so by changes $C_{\text{CH}_3\text{OCH}_3}$ than by $C_{\text{H}_2\text{O}}$.

- One more example. The reaction $2A + B \rightarrow C$ is 1st order in B and -2nd order in A. What is the rate law?

$$-r_A = k \frac{C_B}{C_A^2}$$

- Sometimes, the rate doesn't follow a power law model. For example, the rate law for Fogler problem 5-14_B for the gas phase dehydration of butanol over a silica-alumina catalyst is $-r_A = \frac{kP_A}{(1+K_C P_A)^2}$, where P_A is the partial pressure of A (A = butanol), and K_C is a constant (specifically, it is the adsorption equilibrium constant for butanol on the silica-alumina catalyst). P_A is related to C_A through the ideal gas law: $C_A = \frac{P_A}{RT}$. In this case, the rate law

that you'd use in your mole balance equation is $-r_A = \frac{kC_A RT}{(1+K_A C_A RT)^2} = \frac{k' C_A}{(1+K_P C_A)^2}$ (where the relationship between K_P and K_C is given in Equations 61 and 64 in the Reference Book).

- Some tips:

- Don't overthink construction of the rate law. Just do what the problem is telling you.
- Always write the rate law that you're working with as part of your problem solution. That way, if you do misinterpret what the rate law should be, it can be easily seen.

Let's look at another example. The reaction $A + 2B \rightleftharpoons 3C$ is reversible and exhibits elementary kinetics. What is the rate law?

$$-r_A = k_{Af} C_A C_B^2 - k_{Ar} C_C^3$$

Key definition: $K_C = \frac{k_f}{k_r}$ where K_C is the equilibrium constant

Hence,
$$-r_A = k_{Af} \left(C_A C_B^2 - \frac{C_C^3}{K_C} \right)$$

An aside: What is the difference between a reversible reaction and a reaction in equilibrium?

- In a reversible reaction, both the forward and reverse rates are appreciable. (Whereas for an irreversible reaction, only the forward rate is appreciable; the reverse rate is negligible.)
- An equilibrium reaction represents the special case where the forward and reverse rates are equal to each other (and hence the net rate is equal to 0).
- More on this in a couple of lectures.

Finally, we need to discuss relative rates. The rate expressions for different species in a chemical reaction are related to each other via stoichiometry. For example:

- For the reaction $2A + B \rightarrow C$, we said $-r_A = k_A \frac{C_B}{C_A^2}$. What are the units on $-r_A$?

units on rate in general are $\frac{\text{mol}}{\text{V} \cdot \text{time}}$

units on $-r_A$ are $\frac{\text{mol A}}{\text{V} \cdot \text{time}}$

- What is the relationship between $-r_A$ and $-r_B$?

for every 2 mol A that reacts, 1 mol B reacts

$$\frac{r_A}{-2} = \frac{r_B}{-1} \quad \therefore \quad r_B = \frac{1}{2} r_A$$

- Key equation: Equation 3-1 in Fogler and Equation 45 in the Reference Book: For the reaction $aA + bB \rightarrow cC + dD$:

$$\frac{r_A}{-a} = \frac{r_B}{-b} = \frac{r_C}{c} = \frac{r_D}{d}$$

- Using this equation, what is r_C for the elementary, reversible reaction $A + 2B \rightleftharpoons 3C$, given

$$-r_A = k_{A,f} C_A C_B^2 - k_{A,r} C_C^3?$$

$$r_C = -3 r_A$$

$$\frac{r_A}{-1} = \frac{r_C}{3} \quad , \quad r_C = -3 r_A$$